Journal of Management
Vol. 42 No. 6, September 2016 1557–1587
DOI: 10.1177/0149206313511115

An Investigation of Interpersonal Ties in Interorganizational Exchanges in Emerging Markets: A Boundary-Spanning Perspective

Ying Huang
University of Massachusetts–Lowell
Yadong Luo
University of Miami
Sun Yat-Sen University
Yi Liu
Shanghai Jiao Tong University
Qian Yang
South China University of Technology

This study examines personal ties between boundary-spanning personnel in interorganizational exchanges in emerging markets. Drawing on social embeddedness theory and boundary spanning theory, we propose that strong ties between boundary spanners may benefit exchange parties in their interorganizational relationships through two heightened boundary-spanning behaviors—information processing and external representation. The results from 225 manufacturer–distributor dyads in China indicate that interpersonal ties at the higher levels (between top executives) and at the lower levels (between salespersons and individual buyers) are both positively associated with relationship quality of the buyer–supplier relationship through dyadic boundary-spanning behaviors. Between two levels of interpersonal ties, ties at the lower levels exhibit a stronger association with relationship quality than do ties at the higher levels. Furthermore, the positive effects of interpersonal ties on conflict resolution and cooperation are amplified when both levels of ties are strong in the focal relationship.

Acknowledgments: This article was accepted under the editorship of Deborah E. Rupp. The authors would like to thank Associate Editor Jeremy Short, and three anonymous Journal of Management reviewers for their helpful comments on earlier versions of the article. All errors remain the responsibility of the authors. The authors gratefully acknowledge the financial support by NSFC grants (71172128; 71132006; 71090403) of P.R. China.

Corresponding author: Yadong Luo, School of Business Administration, University of Miami, Jenkins Building, 417D, Coral Gables, FL 33124-9145, USA.

E-mail: yadong@miami.edu

Keywords: *boundary spanner; emerging market; interorganizational relationships; ties*

During the past three decades, research interest in the role of ties in driving superior performance has escalated (e.g., Park & Luo, 2000; Uzzi, 1997; Zhang, Soh, & Wong, 2010). Particularly for firms in emerging markets, ties have become a strategic tool that helps firms to (a) secure key and scarce resources (Luo, 2007), (b) facilitate business transactions through informal social mechanisms (Granovetter, 1985), and (c) navigate changes in institutional and market environments where there is relative underdevelopment of a legal framework (Peng & Heath, 1996). While the importance of ties has been widely recognized, there is still a lack of understanding concerning ties in interorganizational exchanges, and about whether (and how) interpersonal ties between individuals can benefit interorganizational relationships in terms of the quality of exchange relationships. As is correctly identified by Palmatier, Dant, and Grewal (2007: 173), extant interorganizational exchange research involving ties or networks is relatively limited. To address the research gap, this study examines interpersonal ties between boundary-spanning personnel in interorganizational exchanges of emerging markets, and the mechanisms through which ties at the interpersonal level are associated with relationship quality at the interorganizational level.

Boundary-spanning personnel are key representatives who engage in various activities on the boundary of an organization, and who have two primary boundary-spanning roles: information processing and external representation (Aldrich & Herker, 1977). Through these roles, boundary spanners not only support the exchange of information with the external environment, but also facilitate organizational responses to environmental influences (Tushman & Scanlan, 1981). As a platform for evaluating these roles, social embeddedness theory suggests that economic actions are embedded in interpersonal ties and relations (Granovetter, 1973). In many emerging markets, the use of personal ties for business relationships is an indispensable phenomenon (Wright, Filatotchev, Hoskisson, & Peng, 2005). Boundary spanners who are involved in business decisions and operations are likely to be affected by their personal ties with each other. Strong ties may help boundary spanners perform boundary-spanning roles in two ways: (a) serving as a robust base for connecting and sharing information (i.e., information processing) with other boundary spanners, and (b) acting as a relationship lubricant for effective cooperation and problem solving (i.e., external representation). These enhanced boundary-spanning behaviors, in turn, benefit the organizations that boundary spanners represent, and nurture favorable relationships with their exchange partners.

Furthermore, boundary-spanning personnel include boundary spanners at a range of levels within the corporate structure (Aldrich & Herker, 1977). Previous studies on personal ties between boundary spanners have concentrated on ties between top executives, for example, managerial ties in Park and Luo (2001) and in Peng and Luo (2000). Though boundary spanners do include executives at the higher levels (Westphal, Boivie, & Chng, 2006), they also include salespersons and individual buyers at the lower levels (Su, Yang, Zhuang, Zhou, & Dou, 2009), yet the role of lower-level boundary spanners in interorganizational exchanges has been largely overlooked in the literature. Addressing the fact that, in the context of interorganizational exchanges, interpersonal ties do exist at both higher and lower tiers (and, probably, at other tiers), this study investigates interpersonal ties between boundary spanners both at the higher levels (between top executives) and at the lower levels (between salespersons and individual buyers).

In examining interpersonal ties between boundary spanners in interorganizational exchanges in emerging markets, this study seeks to make three contributions. The first contribution is theoretical. By adding a boundary-spanning lens to social embeddedness theory, this study introduces a boundary-spanning mechanism that explains how interpersonal ties are related to relationship quality of exchange relationships in emerging markets. The second contribution is related to the empirical testing. By approaching interpersonal ties at the higher and the lower levels, this study pursues answers to three questions: (a) whether (and how) interpersonal ties between boundary-spanning personnel (i.e., micro level) are associated with relationship quality of interorganizational exchanges (i.e., macro level), (b) whether (and how) ties at the higher levels differ from ties at the lower levels in their associations with relationship quality, and (c) whether ties at the higher and lower levels are complementary or substitutive. The third contribution is methodological. Departing from previous studies that typically use data collected from one side to examine dyadic exchange relationships, this study uses paired dyadic data to explore interpersonal ties and interorganizational relationships, which is an effort reflecting dyadic situations in interorganizational exchanges. Accordingly, the unit of analysis of this study is the dyad. Our empirical context is China, a leading emerging economy and an important source of global manufacturing. This setting is appropriate because of the social embeddedness of Chinese business relations and the pervasive use of personal ties in economic transactions.

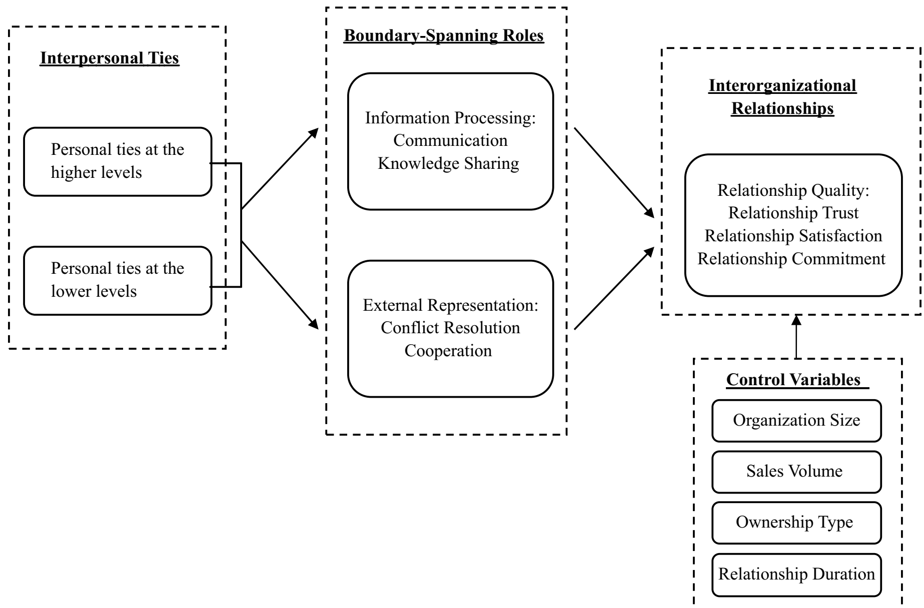
Theoretical Development

A Socially Embedded Ties Perspective

The notion of ties originated in anthropology and sociology. Beginning in the 1970s, ties were introduced to organizational studies, and attention from organizational scholars on ties research continues to increase (Granovetter, 1973; Uzzi, 1997). Scholars who examine the characteristics of interpersonal ties have defined ties as the ongoing reciprocity and friendship between individuals (Luo, 2007), or the close relationship, liking for, and friendship generated from social activities and business transactions (Adobor, 2006; Marsden & Campbell, 1984). Consistent with Granovetter (1973), interpersonal ties discussed in this article are assumed to be dyadic, positive, and symmetric, and are subjectively perceived based on social interactions.

Many theories have been adopted to explain the governance mechanisms of personal ties, including social embeddedness theory, social capital theory, social network theory, the resource-based view, transaction cost economics, and institutional theory. Among those, social embeddedness theory, social capital theory, and social network theory all belong to economic sociology and have been widely used by researchers due to their emphasis on social governance in explaining personal ties. Among these three theories, however, social embeddedness theory is the fundamental theory because, based on the central concept that social embeddedness refers to the influence of prior ties on subsequent economic behaviors, both social network theory and social capital theory are derived from social embeddedness

Figure 1
A Boundary-Spanning Perspective Toward Interorganizational Relationships



theory (Barden & Mitchell, 2007). Accordingly, we have selected social embeddedness theory as the overall theoretical framework for explaining the link between interpersonal ties and subsequent performance outcomes of interorganizational relationships (i.e., relationship quality) in interorganizational exchanges (see Figure 1).

Social embeddedness theory (Granovetter, 1973, 1985; Uzzi, 1997) proposes that economic actions are progressively embedded in a structure of ties and personal relations. The micro-level ties of individuals affect not only the behavioral capacity of individuals, but also the macro-level performance of the organizations to which the individuals belong (Uzzi, 1997). Thus, an analysis of ties provides a necessary and fruitful vehicle for understanding the micro–macro link in interorganizational exchanges and relationships. In interorganizational exchanges, as in other business transactions, the transactions are initiated and carried out transactions by individuals (Tsang, 1998). An individual's ties, although remaining as the "personal property" of the individual, can affect his or her organization (Zhang & Zhang, 2006). Through the appropriate establishment of interactions and obligations, a person's ties can effectively assist in acquisition of the resources and skills necessary to accomplish organizational tasks, to upgrade organizational productivity, and to expand business transactions (Park & Luo, 2001). Following social embeddedness logic, we argue that interorganizational exchange activities are embedded in the social relations of the personnel who are involved in the exchanges. Accordingly, we expect that strong ties formed between personnel, particularly between boundary spanners who are involved in the exchanges, help exchange organizations to achieve superior performance from the exchange relationships. While the strength of interpersonal ties is related to the amount of time spent together and the intimacy between individuals (Granovetter, 1985), the performance of exchange relationships is gauged by a composite construct called relationship quality because, according to relationship marketing literature, relationship quality "provides the most insight into exchange performance" in interorganizational exchanges (Palmatier, Dant, Grewal, & Evans, 2006: 136).

Although social embeddedness theory provides an overall framework for revealing the influences of boundary-spanner ties on relationship quality of exchange relationships, the underlying mechanism of such influences is not clear. With regard to interorganizational exchanges, we argue that the influence of individuals on organizations must function through individual behaviors and interactions. In other words, we suggest that the link between boundary-spanner ties and relationship quality be mediated by boundary-spanning behaviors in exchange activities, and we use boundary-spanning theory to explain this mediation.

Adding the Boundary-Spanning Lens

Organizations depend on external environments for critical resources and business opportunities in order to survive and grow (Stock, 2006). According to boundary-spanning theory (Aldrich & Herker, 1977), organizations rely on boundary spanners to ensure that the social and economic exchanges between an organization and its external environment are executed smoothly, and the organization is thus protected from disruptive external environmental forces. To achieve these goals, boundary spanners perform two major roles—information processing and external representation (Aldrich & Herker, 1977).

When performing the role of information processing, boundary spanners receive information from the external environment, which they decode, filter, and translate before passing it to relevant internal users (Aldrich & Herker, 1977; Tushman & Scanlan, 1981). Boundary spanners also share appropriate internal information with external organizations. Facilitating two-way information flow and sharing, boundary spanners serve as a bridge between the organization and the external environment (Stock, 2006). In interorganizational exchanges, communication is an information transmission process, serving as the bond that holds exchange parties together (Mohr & Spekman, 1994). Similarly, knowledge sharing, as one of the most efficacious methods through which organizations acquire knowledge from each other (Foy, 1999), is a learning facilitation process that helps develop favorable relationships between exchange parties (Griffith, Zeybek, & O'Brien, 2001). As such, we identify communication and knowledge sharing as two important information processing behaviors of boundary spanners engaged in interorganizational exchanges.

Boundary spanners carry out their role as external representative for their organizations by facilitating resource sharing, conveying perceptions and expectations, and providing coordinated assistance to the external environment (Aldrich & Herker, 1977). Through boundary spanners, organizations connect with each other and induce collaborative behaviors in order to achieve mutual goals (Zollo, Reuer, & Singh, 2002). In interorganizational exchanges, cooperation refers to the similar or complementary coordinated actions taken by exchange parties to reach joint outcomes (Anderson & Narus, 1990). On one hand, with the awareness that exchange organizations are increasingly interdependent, cooperation has become increasingly important, and on the other hand, unexpected situations occur and conflicts arise, which require boundary spanners to serve as problem-solving conduits as well, negotiating and resolving issues through rational persuasions and joint actions (Friedman & Podolny, 1992). Accordingly, we view cooperation and conflict resolution as two important external representation behaviors that boundary spanners perform.

Hypotheses

Ties are ubiquitous in emerging markets (e.g., *guanxi* in China; *blat* in Russia; *compadre* in Latin America), which supports the proposition that socially embedded economic transactions (Granovetter, 1973) are particularly relevant for emerging markets. In China, for instance, personal ties are viewed as the lifeblood of the business economy (Xin & Pearce, 1996), and the utilization of ties helps to define the success of organizations (Li, Poppo, & Zhou, 2008; Peng & Luo, 2000). Extending social embeddedness logic to interorganizational exchanges, we argue that exchange transactions and relationships are embedded in interpersonal ties of boundary spanners. In emerging markets such as China, interpersonal ties often develop *ex ante* of the formal exchange relationships, and trust is typically established among individuals or organizations with prior shared experience or connections. In emerging markets, interpersonal ties can serve as informal conduits through which exchange personnel reduce psychological distance (Adobor, 2006) and build trust (Yang, 1994) when performing their boundary-spanning roles, thus benefitting exchange relationships. The congenial atmosphere that results from informal ties between boundary spanners also helps facilitate the formation of a positive psychological evaluation of the relationship, leading the exchange organization to believe that the relationship with the other party is satisfying and that a long-term relationship is beneficial for accomplishing tasks and achieving mutual goals (Mohr & Spekman, 1994).

Emerging markets are often characterized by less transparent information channels and weak institutional systems; thus, individuals and organizations have traditionally relied on social relations for information acquisition, contract enforcement, and rights protection (Wright et al., 2005). Strong ties help boundary spanners perform information processing roles in exchanges by providing a robust and informal channel for smooth communication and effective transfer of knowledge (Uzzi, 1997)—two critical forces in determining favorable exchange relationships (Cheung, Myers, & Mentzer, 2011; Palmatier et al., 2006) and superior performance in emerging markets (Su et al., 2009). Bilateral communication and sharing of intelligence enable exchange parties to increase operational efficiency (Griffith et al., 2001), enhance market capabilities (Kotabe, Martin, & Domoto, 2003), and be more responsive to market and industry changes (Gu, Hung, & Tse, 2008). As well, open communication and information sharing cause exchange parties to feel that their partners are honest, sincere, and well-intentioned, which leads to shared trust and confidence in the relationship.

Strong ties between boundary spanners also serve as a relationship lubricant in emerging markets (Park & Luo, 2001), facilitating cooperation and conflict resolution in exchanges. The friendship and obligation inherent in strong personal ties will encourage reciprocity (Granovetter, 1985), thereby inducing cooperative behaviors. In emerging markets with scarce resources, cooperation enables exchange parties to pool resources in order to exploit market opportunities (Gu et al., 2008). Mutual cooperation also increases understanding between exchange organizations, leading both to believe that their partners are applying sincere effort to fulfilling their obligations, and that each party cares about the interests and goals of the other (Granovetter, 1985). Because conflicts between exchange parties may arise from different economic objectives and expectations, as well as from inappropriate utilization and allocation of resources (Zollo et al., 2002), if conflicts were to occur in exchange relationships, strong ties between boundary spanners can serve as informal conduits for conflict resolution (Su et al., 2009). In emerging markets where legal protection and contract enforcement are often weak, organizations may rely on informal social tools for effective conflict resolution (Su et al., 2009). Collective conflict resolution behaviors help identify a satisfying solution that removes misunderstandings, recovers trust, and retains commitment (Anderson & Narus, 1990; Mohr & Spekman, 1994). In emerging markets, therefore, we expect that boundary-spanning behaviors serve as mediators between interpersonal ties and relationship quality in interorganizational exchanges.

Hypothesis 1 (H1): Interpersonal ties between boundary spanners are positively associated with relationship quality between exchange organizations through (a) boundary spanners' information processing behaviors (i.e., communication and knowledge sharing) and (b) boundary spanners' external representation behaviors (i.e., conflict resolution and cooperation).

After establishing the positive link between boundary-spanner ties and relationship quality, we further disentangle interpersonal ties and distinguish their effects in interorganizational exchanges. As mentioned, boundary spanners include top managers at the higher levels (i.e., top executives) as well as employees at the lower levels (i.e., individual salespersons and buyers; Aldrich & Herker, 1977). For organizations in emerging markets, ties between boundary spanners may exist simultaneously at both levels. Extant research, however, has focused on ties at the higher levels (also called managerial ties) but has neglected ties at the lower levels. We argue that, in interorganizational exchanges, boundary-spanner ties at the lower levels may be more important because the foci of individual salespersons and buyers, and their involvement in daily operations and exchange processes, may cause them to play a more critical role in facilitating exchange transactions than that of top executives.

As the chief helmsmen of organizations, top executives are endowed with decision-making power at the strategic levels (Park & Luo, 2001), and typically bring to the focal exchange relationship a familiarity with and mastery of a range of information and knowledge related to organizational tasks such as developing strategic plans, setting organizational orientation, and choosing target markets (Westphal et al., 2006). In emerging markets, however, top executives may face additional challenges. For instance, organizations in China must compete for scarce resources (e.g., bank loans or government subsidies), so that top executives need to cultivate ties with governmental officials (also called political ties; Sheng, Zhou, & Li, 2011). Due to their focus on government connections, top executives in emerging markets may be less involved in exchange relationships, so that ties between them may have less focus and impact on the operational issues in interorganizational exchanges.

In contrast, salespersons and buyers are charged with daily duties and are engaged in concrete business processes and routine affairs at the operational levels. Personal ties formed between salespersons and buyers form a strong basis to communicate effectively and to share valuable information about customers and markets. In interorganizational exchanges, salespersons and buyers are often the personnel who encounter specific conflicts and coordination problems during interactions (Anderson & Narus, 1990). Due to their familiarity with the background, nature, and possible results of conflicts, salespersons and buyers are better equipped to accurately pinpoint the root causes of the problems, and the strong ties between them will enable effective application of appropriate solutions and serve as a call for cooperation. Taken together, we expect that lower-level ties are more effective than higher-level ties for facilitating information processing and external representation, because these boundary-spanning behaviors in interorganizational exchanges are mainly at the operational level. Specifically, we postulate,

> *Hypothesis 2 (H2)*: Compared with interpersonal ties at the higher levels, interpersonal ties at the lower levels have a stronger association with relationship quality through (a) information processing behaviors (i.e., communication and knowledge sharing) and (b) external representation behaviors (i.e., conflict resolution and cooperation).

Once there is an understanding that interpersonal ties exist at both the higher and the lower levels across exchange parties, the next question is, naturally, whether higher-level ties and lower-level ties are complementary or substitutive in interorganizational exchanges. Because we expect that lower-level ties exert stronger influences than higher-level ties on mutual communication, knowledge sharing, conflict resolution, and cooperation, the question is, essentially, whether higher-level ties strengthen or weaken the association between lower-level ties and boundary-spanning behaviors. Based on the notion of the socially embedded economy (Granovetter, 1973), we expect that in emerging markets, two parallel interpersonal ties are complementary for promoting boundary-spanning behaviors in interorganizational exchanges.

In many emerging markets that are transitioning to a more developed economy, a typical problem for organizations is the lack of market information and business intelligence (Luo, 2007). The information scarcity is likely due to the relative underdevelopment of specialized intermediaries such as governmental agencies that collect and disseminate information. Thus, organizations in emerging markets rely heavily on informal ties for information exchange and acquisition (Park & Luo, 2001). When an organization's top executives and lower-level employees both have sound personal ties with their counterparts at the other organization, the result, for both organizations, is a wider range of business information that includes both strategic and operational intelligence that can be shared (Kotabe et al., 2003). Furthermore, lower-level boundary spanners may regard strong ties between top executives as a signal of approval or encouragement, and therefore be more open to communication and knowledge sharing. As a result, we expect a higher level of information processing behaviors when strong ties exist at both the higher and the lower levels.

As well, organizations in emerging markets are likely to face another barrier that is a consequence of the relative underdevelopment of a legal framework for protecting property rights and enforcing contracts (Peng & Heath, 1996). The legal and institutional constraints in emerging markets may pose severe problems when conflicts occur and relationships deteriorate. In this situation, interpersonal ties may serve as a relationship lubricant for resolving conflicts (Park & Luo, 2001) and an informal conduit for inducing cooperation between exchange parties, helping organizations redress deficiencies of legal and institutional environments (Su et al., 2009). In an exchange relationship where strong ties exist at both the higher and lower levels, we may observe a decreased degree of conflict and an increased degree of cooperation because salespersons and buyers may view strong ties between executives as a call for dedication and collaboration. However, when conflicts do occur at the lower levels, they are usually resolved quickly. If the issues are critical and strategic (e.g., incongruent organizational objectives), the conflicts may escalate, reaching the top executives in the search for solutions. In these cases, top executives with strong ties are more likely to work out satisfactory solutions for both parties (e.g., realign objectives and expectations). Thus, we anticipate a higher degree of external representation behaviors when strong ties exist at both the higher and the lower levels. Consequently, we predict,

> *Hypothesis 3 (H3)*: There exists a complementary effect between interpersonal ties at the higher levels and interpersonal ties at the lower levels, such that ties at the lower levels have a stronger association with (a) information processing behaviors (i.e., communication and knowledge sharing) and (b) external representation behaviors (i.e., conflict resolution and cooperation) when ties at the higher levels are strong.

Research Method

Sampling and Data Collection

This study uses matched survey data from both the manufacturer (supplier) and the distributor (buyer) sides, in the context of the Chinese household appliances industry. China has become the world's second largest economy, and for many years has had a steady gross domestic product growth of 9%. The home appliance industry represents one of the earliest and most developed industries in China. It is highly market-oriented, and includes both world-class manufacturers (e.g., Haier, TCL, Changhong) and world-class distributors (e.g., Walmart, SuNing, Gome). Our focus on the single industry helps to rule out between-industry differences in buyer–supplier practices in China.

Our data collection consisted of three stages. First, before designing the survey questionnaire, we selected seven manufacturers and four distributors as potential subjects.[1] We conducted unstructured personal interviews with 11 senior managers in order to develop an understanding of interpersonal ties and boundary-spanning interactions in the manufacturer–distributor relationships. These senior-manager interviewees indicated that, because they hold a unique position in the middle of the organizational structure, they work closely (and daily) with their supervisors (i.e., top executives) and supervisees (i.e., individual salespersons or buyers), and therefore were familiar with interpersonal ties and interorganizational interactions at both the higher and the lower levels. Overall, the managers confirmed that personal ties exist at both levels. For example, one interviewee mentioned, "My sales representative brings gifts and sends text messages for the buyers' birthdays and holidays; this is important to our relationship with their company"; and another interviewee revealed, "Some collaboration ideas were brought up between our bosses during social gatherings after work."

Second, we developed paired questionnaires in English for both manufacturers and distributors. The English version was translated into Chinese and then back-translated into English, with recognition of the cross-cultural distinctions in the connotation of the equivalent constructs. The back-translated English version was then checked against the original English version and a number of questions were reworded to improve the accuracy of the translation. To ensure the clarity of the questionnaire, we deployed a research team and conducted a pilot test with 16 manufacturer–distributor dyads, using semistructured interviews to check each of the items. Based on the feedback, final refinements were made.

Finally, we began dyadic data collection with the distributors. A sampling frame that contains information on 900 nationwide distributors was provided by the Haier Group, the largest and most prestigious household appliance manufacturer in China (and third largest in the world). Haier Group requested that its distributors participate in our survey, communicating that request to them through letters and phone calls. We followed up by mailing coded surveys to each of the distributors. At each distributor location, contact persons were asked to select a major manufacturer (other than Haier) and to describe their relationship with that manufacturer. In the survey questionnaire, a major manufacturer was defined as the manufacturer from which the distributors purchased the largest volume during the last three months. In cases where Haier was the manufacturer, the distributor was directed to choose the manufacturer with the second-largest purchase volume during the period. To encourage participation, we offered to provide a report of the research results, if desired. A statement of confidentiality was also included in the cover letter of the questionnaire. After three rounds of reminders (by phone, by initial and follow-up emails, and in person), 314 questionnaires were returned, of which 251 were complete. The paired questionnaires were then sent to designated manufacturers that had been selected by the distributors, asking the manufacturers to assess their relationships with the distributors. Using encouragement and reminder measures identical to those of the first questionnaire, 225 completed questionnaires were received from 251 manufacturers—an overall response rate of 25%. Geographically, manufacturers and distributors were located evenly across the country.

To ensure that the respondents were capable of answering questions, we requested that contact persons be senior managers who deal directly with the other party, and who are fully responsible for the organizational relationship. These senior managers are typically sales managers or purchasing managers (73.6% for manufacturers, and 74.5% for distributors) who supervise sales/purchasing personnel and who report to top executives. Our prior interviews had confirmed their capability of answering questions related to boundary spanners at the higher and lower levels. To further validate the respondents' qualifications, we included two items that help to assess the knowledge level of the respondents. The first question asked the informants' specific tenure in their current positions—which yielded results of an average 3.5 years in the position for manufacturers, and 4.7 years for distributors. The second item was intended to determine the extent of the informants' familiarity with the focal relationship. On a 5-point Likert-type scale, we obtained a mean of 4.23 (SD = 0.56) for manufacturers and 4.25 (SD = 0.62) for distributors. Finally, we guaranteed full anonymity for all respondents in order to minimize social desirability bias.

The methodological design supporting bilateral data provides our study with the advantage of a reduction in the common method variance bias that is typically associated with unilateral data. Following the steps recommended by Podsakoff and Organ (1986), we conducted a global factor analysis on items subordinate to all prediction and criterion variables of both sides. No single factor emerges from the analysis, and no one factor accounts for most of the covariance for all variables. Furthermore, following Podsakoff, Mackenzie, Podsakoff, and Lee (2003), we added a general method factor to the measurement model. The results (χ^2 = 4369.36, df = 594, $p < .001$; comparative fit index [CFI] = .46, normed fit index [NFI] = .43, root mean square error of approximation [RMSEA] = .17) indicate that the model fit deteriorates, suggesting that common method variance bias is unlikely in this study. As a final analytical concern, we examined the nonresponse bias by randomly choosing 50 firms from each side that either did not complete or did not return their questionnaires, and we collected information for comparison *t* tests, focusing on firm attributes (size, sales volume, location, and ownership). None of the results were significant, demonstrating the absence of nonresponse bias.

Measures and Validation

The constructs in this study are derived from the literature and adapted to our specific research context. Except for relationship duration (a single item that was collected only from the distributors), all constructs are measured using a 7-point Likert-type scale, with end points of *strongly disagree* and *strongly agree* (see Table 1 for constructs and items, and Table 2 for descriptive statistics). Because of the dyadic nature of interpersonal ties and interorganizational relationships, all constructs in this study (expect control variables) are treated as dyadic constructs.

Interpersonal ties. A primary dimension of ties is the strength of ties characterized by the amount of time, emotional intensity, and reciprocal services of social actors in the exchange (Granovetter, 1985). In accordance with Adobor (2006), Granovetter (1973), Marsden and Campbell (1984), and Uzzi (1997), we define dyadic interpersonal ties as a close and symmetric personal relationship generated from social and leisure activities between boundary spanners. Based on this definition and on the work of Ambler, Styles, and Wang (1999), two items were developed to identify the personal ties between boundary spanners at the higher levels (between top executives) and at the lower levels (between salespersons and buyers), measuring degrees of boundary spanners' involvement in socialization and personal favors.

Information processing roles. Boundary spanners' information processing behaviors include mutual communication and knowledge sharing between boundary spanners. Communication represents the process of valuable business information exchange between exchange parties (Mohr & Spekman, 1994). Knowledge sharing denotes a timely two-way conveyance of knowledge, including knowledge about products, competitive forces, markets, and competitors. Four items extracted from Mohr and Spekman and five items from Griffith et al. (2001) were used to assess the two respective constructs.

External representation roles. Boundary spanners' external representation behaviors include mutual conflict resolution and cooperation. Three items from Mohr and Spekman (1994) were used to measure conflict resolution, which is the manner or technique used by exchange members to solve their manifest conflicts. Identical to the constructs of Anderson and Narus (1990), cooperation in our study indicates complementary coordinated actions taken by exchange members in order to achieve mutual outcomes or profits, with expected reciprocation in various functions over time. With reference to recent studies on the content of cooperation in strategic management, we developed five items to capture the essence.

Relationship quality. Following relationship marketing literature, we used perceived relationship quality to gauge performance outcomes of interorganizational relationships. Relationship quality is a multidimensional construct capturing the different but related facets

Table 1
Construct Reliability and Validity

	Cronbach α			Factor Loading			CR			AVE			R_{wg}
	D	M	DM	D	M	DM	D	M	DM	D	M	DM	
Interpersonal ties at the higher level (between top executives)	.73	.60	.62				.84	.88	.84	.72	.79	.72	.85
TM1: Leaders of both sides always invite each other to participate in various activities for socialization.				.85	.89	.85							
TM2: Our leaders and the leaders of our partner call on each other.				.85	.89	.85							
Interpersonal ties at the lower level (between salespersons and buyers)	.66	.64	.63				.85	.85	.84	.74	.74	.72	.82
SB1: Staffs from both sides have good personal relationships.				.86	.86	.85							
SB2: Staffs from both sides often spend their spare time together in social activities.				.86	.86	.85							
Communication	.94	.92	.95				.96	.94	.96	.85	.81	.87	.93
C1: We inform each other in advance of changing needs.				.90	.89	.91							
C2: We share proprietary information with each other.				.93	.87	.93							
C3: Any information that might help the partner's side will be provided.				.94	.92	.94							
C4: We keep each other informed about events or changes that may affect the other side.				.92	.92	.94							
Knowledge sharing	.86	.88	.89				.90	.92	.92	.65	.69	.70	.94
K1: We have provided (learned) a great deal of knowledge about substitute products to (from) the supplier (buyer).				.75	.75	.77							
K2: We have provided (learned) a great deal of knowledge about competitive advantages to (from) the supplier (buyer).				.86	.89	.89							
K3: We have provided (learned) a great deal of knowledge about market potential to (from) the supplier (buyer).				.85	.88	.88							
K4: We have provided (learned) a great deal of knowledge about competitors to (from) the supplier (buyer).				.83	.86	.85							
K5: We have provided (learned) a great deal of knowledge about their competitors to (from) the supplier (buyer).				.74	.76	.79							
Conflict resolution	.88	.92	.91				.92	.95	.94	.80	.86	.85	.91

(continued)

Table 1 (continued)

	Cronbach α			Factor Loading			CR			AVE			R_{wg}
	D	M	DM	D	M	DM	D	M	DM	D	M	DM	
CF1: When conflict occurs, we exchange complete and accurate information with the supplier (buyer) in order to smooth over the problem.				.90	.92	.92							
CF2: When conflict occurs, persuasive attempts are used by either side, including ignoring the differences and emphasizing the common interests between the supplier (buyer).				.92	.94	.93							
CF3: When conflict occurs, both sides try their best to reach a joint solution to the problem.				.86	.92	.91							
Cooperation	.92	.91	.93				.94	.94	.95	.77	.75	.78	.92
CO1: We have established an effective team with the supplier (buyer).				.88	.82	.87							
CO2: Both sides send a number of technical and managerial personnel to the other side.				.79	.77	.79							
CO3: Information about products, technology, and market are transferred between us.				.92	.91	.93							
CO4: The supplier (buyer) can obtain relevant knowledge through visiting and observing our firm.				.90	.91	.91							
CO5: We conduct a certain amount of shared training for salespersons with the supplier (buyer).				.88	.90	.90							
Relationship trust	.91	.91	.92				.93	.93	.94	.73	.74	.76	.95
RT1: We believe that the supplier (buyer) will not make excessive requests of us.				.84	.78	.82							
RT2: We believe in the supplier (buyer) as being sincere.				.82	.88	.87							
RT3: We can have confidence that the supplier's (buyer's) future decisions and actions will not adversely affect us.				.86	.90	.90							
RT4: When making important decisions, the supplier (buyer) cares about our welfare or interests.				.87	.88	.89							
RT5: When it comes to things that are important to us, we can depend on the supplier's (buyer's) support.				.87	.86	.87							

(continued)

Table 1 (continued)

	Cronbach α			Factor Loading			CR			AVE			R_{wg}
	D	M	DM	D	M	DM	D	M	DM	D	M	DM	
Relationship satisfaction	.93	.91	.93				.95	.93	.95	.79	.74	.78	.95
RS1: The working relationship between our firm and the supplier (buyer) is characterized by feelings of friendliness.				.89	.87	.90							
RS2: The supplier (buyer) expresses criticism tactfully.				.88	.80	.86							
RS3: Interactions between our firm and the supplier (buyer) are characterized by mutual respect.				.92	.89	.91							
RS4: The supplier (buyer) never leaves us in the dark about things we ought to know.				.93	.89	.91							
RS5: The supplier (buyer) always explains to us the reasons for its company policies.				.83	.84	.84							
Relationship commitment	.93	.91	.92				.95	.93	.94	.77	.73	.76	.94
RC1: We intend to continue working with the supplier (buyer) because we feel as if they are "part of family."				.87	.76	.84							
RC2: We would not replace the supplier (buyer), even if another supplier (buyer) made a better offer.				.84	.86	.86							
RC3: Given the same business philosophy as the supplier (buyer), we feel we ought to continue our relationship with the supplier (buyer).				.90	.89	.90							
RC4: We have a strong sense of loyalty to the supplier (buyer), so we continue to work with them.				.89	.91	.89							
RC5: Given all the things we have done with the supplier (buyer) over the years, we feel we ought to continue our relationship with the supplier (buyer).				.90	.83	.88							

Organization size: (1) under 100; (2) 100-500; (3) 501-1,500; (4) 1,501-5,000; (5) 5,001-25,000; (6) above 25,000. Ownership type: (1) wholly state-owned; (2) Sino–foreign joint venture; (3) wholly foreign-owned; (4) other.

Sales volume (RMB): (1) less than 10 million; (2) 10-50 million; (3) 50-200 million; (4) 200 million-1 billion; (5) more than 1 billion. Relationship duration: years.

Note: AVE = average variance extracted; CR = composite reliability; D = distributor; DM = dyadic sides; M = manufacturer.

Table 2
Descriptive Statistics and Pearson's Correlation Matrix (N = 225)

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Higher-level ties	5.28	0.90	1.00												
2. Lower-level ties	4.98	1.05	.62**	1.00											
3. Communication	5.56	0.84	.44**	.50**	1.00										
4. Knowledge sharing	5.33	0.76	.34**	.41**	.49**	1.00									
5. Conflict resolution	5.39	0.86	.34**	.30**	.51**	.37**	1.00								
6. Cooperation	5.00	1.10	.41**	.44**	.50**	.47**	.39**	1.00							
7. Relationship trust	5.62	0.71	.16**	.28**	.39**	.47**	.37**	.42**	1.00						
8. Relationship satisfaction	5.58	0.74	.45**	.44**	.60**	.49**	.54**	.55**	.55**	1.00					
9. Relationship commitment	5.42	0.82	.43**	.45**	.31**	.26**	.30**	.34**	.42**	.49**	1.00				
10. Organization size	2.82	1.11	.04	–.03	.10	–.03	.03	–.10	.02	.07	.04	1.00			
11. Sales volume	3.29	1.05	.17	.02	.17	.03	.11*	.05	.08	.11	.08	.67**	1.00		
12. Ownership type	3.33	0.93	.07**	.11*	–.02*	–.01	–.02	–.02	.13*	.06	.23**	–.09	–.13**	1.00	
13. Relationship duration	5.44	2.25	.02	–.07	–.10	–.01	–.07	–.03	.01	–.03	.10	.10	.16**	.02	1.00

*p < .05, two-tailed.
**p < .01, two-tailed.

of a relationship (Palmatier et al., 2006). Three commonly included components of relationship quality are trust, commitment, and satisfaction. Relationship quality is a second-order factor, with its three components as first-order factors. Our questionnaire used five items from Kumar, Scheer, and Steenkamp (1995) to measure relationship trust, and used another five items based on measures initially developed by Geyskens and Steenkamp (2000) to measure relationship satisfaction. Relationship commitment was measured by five items adapted from Anderson and Weitz (1992).

Control variables. Three variables (i.e., organization size, sales volume, and ownership type) that are specific to organizational characteristics and one that is specific to relationship characteristics (i.e., relationship duration) were included in the model. *Organization size* was operationalized by the number of employees. Size is an important indicator of an organization's structural flexibility and bargaining power, and therefore has an impact on organizational behaviors and relationships (see Droge, Claycomb, & Germain, 2003). Generally, large organizations can be expected to exert greater influence on interorganizational relationships than small ones will (Droge et al., 2003). Overall, the manufacturers in our sample have many more employees than do the distributors. Therefore, we use size from the manufacturer side as a control variable. *Sales volume* was measured using total sales from the previous year. In previous studies (e.g., Mohr & Spekman, 1994) sales volume has been used as an objective indicator of a successful relationship, leading us to anticipate that sales volume will likely exert a positive effect on relationship quality. In general, the manufacturers in our sample have a greater sales volume than the distributors. Thus, we include sales volume from the manufacturer side as another control variable. *Ownership type* was coded as a dummy variable, measuring degrees ranging from wholly state-owned to wholly foreign-owned. Ownership type is an important proxy for organizational culture. Li et al. (2008) report that, compared with domestic firms, foreign firms have a competitive disadvantage in terms of performance gains from tie utilization. Thus, it is necessary to include ownership type as a control variable. Because the manufacturers tend to be more powerful than the distributors in the Chinese home appliance industry, we adopt data from the manufacturer side as a control variable. Finally, *relationship duration* was measured by the number of years the manufacturer had been dealing with the distributor. The relationship marketing literature suggests that relationship duration has a positive effect on relationship quality (Kumar et al., 1995; Palmatier et al., 2006). We chose to use the information collected from the distributor in an effort to mitigate common method bias, as the other three control variables were from the manufacturer side.

Using the following steps, we verified the reliability and validity of all constructs. First, an item-to-total correlation examination suggested by Anderson and Gerbing (1988) indicates that there are no deviations from the external consistency (all > 0.4). Second, internal consistency was assessed using Cronbach's alpha. The results for all constructs were greater than .7 (with the exceptions of .62 for personal ties between top executives and .63 for personal ties between salespersons and buyers, because each construct has only two items), which establishes high internal consistency. Furthermore, based on methods from Bagozzi and Yi (1988), the composite reliability (CR) values were calculated, demonstrating high construct reliability (all > .83). Third, all items were subjected to exploratory factor analysis using both orthogonal and oblique rotation to make certain that they have high loadings on the factors for the category they belong to, and that they have low loadings on other factors for both the distributor and the manufacturer data. Fourth, we conducted confirmatory factor analysis (CFA) to assess factor unidimensionality, and the results confirm the unidimensionality of each construct (goodness-of-fit index [GFI] = .90, CFI = .96, RMSEA = .06). In addition, the factor loadings of all items are greater than .7, which indicates construct validity (Bagozzi & Yi, 1988). Fifth, the average variance extracted (AVE) values for all constructs exceed .5, illustrating good convergent validity. Sixth, discriminant validity was assessed by three different methods: (a) Following Fornell and Larcker (1981), the squared correlations between each pair of constructs were compared to the AVE value for each construct, and all squared correlations are less than the AVE values, confirming the acceptable discriminant validity; (b) we calculated the 95% confidence interval of the correlations between any two randomly selected constructs, as recommended by Anderson and Gerbing (1988), and did not find 1.0 included in any interval; (c) we assessed discriminant validity using two-factor CFA models for all possible pairs of constructs. For each pair of constructs, discriminant validity was assessed by comparing the fit between the one-factor model and the two-factor CFA model. In the one-factor model, all measures were forced to load on a single factor. In the two-factor model, each measure was allowed to load only on its respective latent factor. The chi-square values of the two-factor models are significantly lower than those of the one-factor models, $\Delta\chi^2(1) > 3.84$, providing further evidence of discriminant validity between the principal constructs.

Analysis

Results From Structural Equation Model (SEM) Testing

All dyadic constructs are operationalized by taking mean scores of both parties' scores in the dyads, following the aggregation approach in Yan, Simsek, and Lubatkin (2008). To validate this approach, we checked the interrater agreement index (R_{wg}) within the dyad (James, Demaree, & Wolf, 1984). The resulting R_{wg} indices for all constructs in this study range from .82 to .95 (see Table 1), which demonstrates a high degree of perception convergence between the two exchange parties on interpersonal ties, role behaviors, and relationship quality. The scores of R_{wg} for interpersonal ties at the higher and the lower levels are .85 and .82, respectively, confirming Granovetter's (1973) view that dyadic interpersonal ties are positive and symmetric. Thus, we determine that the aggregation approach for obtaining the proxies for dyadic constructs in this study is appropriate.[2] The interaction term is the product of interpersonal ties at both levels; we mean center the indicators of two variables to reduce the multicollinearity. To test the proposed hypotheses, we use a series of structural equation models (see Table 3).

H1 proposes the mediation effects of boundary spanners' mutual information processing and reciprocal external representation behaviors. Using traditional mediation analysis from Baron and Kenny (1986), H1a and H1b are tested by three models: Model 1 tests the direct relationships between interpersonal ties and the relationship quality, Model 2 tests the indirect relationships between interpersonal ties and relationship quality mediated through four behaviors (communication, knowledge sharing, conflict resolution, and cooperation), and Model 3 includes both direct and indirect relationships. As indicated in Table 3, Model 1 reveals the positive and significant associations between interpersonal ties and relationship

Table 3
Hypotheses Testing: Structural Equation Model Results (N = 225)

Estimates							
Model	χ^2	*df*	CFI	GFI	RMSEA	NFI	IFI
Model 1	303.63	219	.98	.90	.04	.92	.98
Model 2	865.73	624	.97	.86	.04	.90	.97
Model 3	864.23	621	.97	.86	.04	.89	.97

Path	Model 1	Model 2	Model 3
Interpersonal ties at the higher level—Relationship quality	0.36** (0.13)		0.18 (0.18)
Interpersonal ties at the lower level—Relationship quality	0.34** (0.09)		0.33 (0.43)
Higher- × lower-level interpersonal ties—Relationship quality	0.08* (0.03)		0.04 (0.05)
Interpersonal ties at the higher level—Communication		0.54** (0.12)	0.51** (0.11)
Interpersonal ties at the higher level—Knowledge sharing		0.44** (0.12)	0.42** (0.12)
Interpersonal ties at the higher level—Conflict resolution		0.71** (0.16)	0.66** (0.16)
Interpersonal ties at the higher level—Cooperation		0.52** (0.15)	0.49** (0.15)
Interpersonal ties at the lower level—Communication		1.20** (0.26)	1.23** (0.27)
Interpersonal ties at the lower level—Knowledge sharing		1.01** (0.25)	1.03** (0.26)
Interpersonal ties at the lower level—Conflict resolution		1.17** (0.29)	1.19** (0.30)
Interpersonal ties at the lower level—Cooperation		1.47** (0.33)	1.51** (0.34)
Higher- × lower-level interpersonal ties—Communication		0.04 (0.05)	0.04 (0.05)
Higher- × lower-level interpersonal ties—Knowledge sharing		0.11† (0.06)	0.10† (0.06)
Higher- × lower-level interpersonal ties—Conflict resolution		0.24** (0.07)	0.23** (0.07)
Higher- × lower-level interpersonal ties—Cooperation		0.27** (0.07)	0.26** (0.07)
Communication—Relationship quality		0.04 (0.08)	–0.15 (0.29)
Knowledge sharing—Relationship quality		0.12† (0.07)	0.14 (0.07)
Conflict resolution—Relationship quality		0.17** (0.05)	0.13* (0.06)
Cooperation—Relationship quality		0.13** (0.04)	0.10* (0.05)
Control variables effects			
Organization size—Relationship quality	0.04 (0.03)	0.05** (0.02)	0.05** (0.03)
Sales volume—Relationship quality	–0.02 (0.04)	–0.02 (0.02)	–0.01 (0.03)
Ownership type—Relationship quality	0.04 (0.03)	0.06*** (0.02)	0.07*** (0.02)
Relationship duration—Relationship quality	0.01 (0.01)	0.01 (0.01)	–0.00 (0.02)

Note: CFI = comparative fit index; GFI = goodness-of-fit index; IFI = incremental fit index; NFI = normed fit index; RMSEA = root mean square error of approximation. Values in parentheses are standard errors.
†p < .1.
*p < .05.
**p < .01.
***p < .001.

quality when only the direct relationships are included, Model 2 lists the results after four mediators are inserted, and, finally, Model 3 shows that the direct paths from interpersonal ties to relationship quality become nonsignificant after the four mediators are included. The fit indices of Models 2 and 3 are similar; however, the fact that the chi-square difference between the two models is minimal ($\Delta\chi^2$ = 1.5, *df* = 3) suggests that Model 2 is superior due to its parsimoniousness. The results in Model 2 show that the paths from interpersonal ties (at both levels) to all boundary-spanning behaviors are positive and significant ($p < .01$), but the paths from boundary-spanning behaviors to relationship quality vary, with knowledge sharing ($p < .1$) and communication ($p > .1$) demonstrating little to no impact, and with conflict resolution ($p < .01$) and cooperation ($p < .01$) exhibiting significant influences on relationship quality. Taken together, these results suggest that, in interorganizational exchanges, the mediation effects of boundary spanners are confirmed for external representation behaviors (i.e., conflict resolution and cooperation) but are not quite confirmed for information processing behaviors (i.e., communication and knowledge sharing). Thus, H1a is not supported and H1b is fully supported.

H2 compares two different levels of interpersonal ties on their associations with relationship quality through boundary-spanning behaviors. Following Baron and Kenny (1986), the overall mediated effect is calculated as the product of path estimates (i.e., a × b). Through communication, the overall effects of higher-level and low-level interpersonal ties on relationship quality are calculated as 0.022 and 0.048, respectively. Through knowledge sharing, the overall effects of higher-level and lower-level interpersonal ties on relationship quality are calculated as 0.053 and 0.121, respectively. Through conflict resolution, the overall effects of higher-level and lower-level interpersonal ties on relationship quality are calculated as 0.121 and 0.199, respectively. Through cooperation, the overall effects of higher-level and lower-level interpersonal ties on relationship quality are calculated as 0.068 and 0.191, respectively. Using chi-square difference tests, we compare the overall effects of two different levels of interpersonal ties by constraining the two paths to be equal, and then releasing the constraint. The results show that lower-level interpersonal ties have a stronger association with relationship quality than do higher-level interpersonal ties. This result is realized through communication ($p < .05$), knowledge sharing ($p < .05$), and cooperation ($p < .01$), but not through conflict resolution ($p > .1$). Therefore, H2a is fully supported while H2b is partially supported.

For interpersonal ties at both upper and lower levels, H3 posits the interaction effects on boundary-spanning behaviors. The results in Model 2 and in Model 3 show that the interaction of two levels of interpersonal ties exerts significant influence on conflict resolution ($p < .01$) and cooperation ($p < .01$), but little on knowledge sharing ($p < .1$) and communication ($p > .1$). These results suggest that strong ties between top executives enhance only the positive effects of personal ties between salespersons and buyers on their behaviors of external representation, but not on their behaviors of information processing. Therefore, H3a is not supported, while H3b is supported.

Results Validation From Bootstrap Testing

We further validate the results using bootstrapping (Efron & Tibshirani, 1993), another method recommended by Bollen and Stine (1990) for the analysis of direct and indirect effects in mediation.[3] First, we test whether boundary-spanning behaviors are mediators between interpersonal ties and relationship quality. Regarding the mediation role of communication, the bias-corrected bootstrap estimates (95% CI) of a1 (0.05, 0.97) and a2 (0.27, 1.00) exclude zero (see Table 4), indicating that both higher-level ties and lower-level ties are positively associated with communication. The estimates of b1 (–4.18, 0.68) include zero, which means that communication is not related to relationship quality. In addition, the

Table 4
Hypothesis Testing: Bootstrap Results ($N = 1{,}000$)

	95% CI							
	Communication		Knowledge Sharing		Conflict Resolution		Cooperation	
Effect	Bootstrap Percentile	Bias-Corrected Bootstrap	Bootstrap Percentile	Bias-Corrected Bootstrap	Bootstrap Percentile	Bias-Corrected Bootstrap	Bootstrap Percentile	Bias-Corrected Bootstrap
Higher-level ties								
a1	(0.06, 0.98)	(0.05, 0.97)	(0.03, 0.77)	(0.10, 0.94)	(0.09, 1.05)	(0.09, 1.07)	(0.02, 0.91)	(0.01, 0.88)
b1	(–1.79, 1.06)	(–4.18, 0.68)	(0.05, 0.63)	(0.04, 0.60)	(0.01, 0.27)	(0.03, 0.28)	(0.06, 0.24)	(0.06, 0.24)
c1’	(–0.42, 1.48)	(–0.14, 4.18)	(0.03, 0.37)	(0.07, 0.41)	(0.02, 0.40)	(0.01, 0.31)	(0.02, 0.30)	(0.04, 0.32)
a1*b1	(–0.79, 0.62)	(–3.31, 0.27)	(0.01, 0.26)	(0.02, 0.38)	(–0.04, 0.14)	(0.03, 0.18)	(0.01, 0.18)	(0.01, 0.18)
Lower-level ties								
a2	(0.27, 1.00)	(0.27, 1.00)	(0.33, 4.24)	(0.36, 4.91)	(0.12, 0.82)	(0.14, 0.82)	(0.42, 1.01)	(0.42, 1.02)
b2	(–1.79, 1.06)	(–4.18, 0.68)	(0.05, 0.63)	(0.04, 0.60)	(0.01, 0.27)	(0.03, 0.28)	(0.06, 0.24)	(0.06, 0.24)
c2’	(–0.35, 1.55)	(–0.22, 2.08)	(–0.03, 1.88)	(0.01, 2.21)	(0.16, 0.50)	(0.14, 0.48)	(0.12, 0.38)	(0.12, 0.39)
a2*b2	(–0.96, 0.60)	(–2.30, 0.35)	(0.04, 1.13)	(0.04, 1.10)	(0.01, 0.13)	(0.01, 0.13)	(0.04, 0.18)	(0.05, 0.19)
Higher × lower interaction								
a3	(–0.14, 0.09)	(–0.17, 0.15)	(–0.04, 0.66)	(–0.03, 0.69)	(–0.02, 0.25)	(0.01, 0.26)	(0.05, 0.36)	(0.04, 0.34)
b3	(–1.79, 1.06)	(–4.18, 0.68)	(0.05, 0.63)	(0.04, 0.60)	(0.01, 0.27)	(0.03, 0.28)	(0.06, 0.24)	(0.06, 0.24)
c3’	(0.01, 0.23)	(–0.02, 0.22)	(–0.01, 0.35)	(0.01, 0.38)	(–0.03, 0.10)	(–0.03, 0.10)	(–0.04, 0.08)	(–0.04, 0.09)
a3*b3	(–0.11, 0.07)	(–0.03, 0.54)	(–0.01, 0.20)	(–0.01, 0.24)	(–0.01, 0.04)	(0.01, 0.05)	(0.01, 0.06)	(0.01, 0.06)

Note: Using the higher-level ties–communication–relationship quality path as an illustration: a1 denotes the effect of higher-level ties on communication; b1 denotes the effect of communication on relationship quality; c1’ denotes the direct effect of higher-level ties on relationship quality; a1*b1 denotes the indirect effect of higher-level ties on relationship quality through communication.

estimates of a1*b1 (–3.31, 0.27) and a2*b2 (–2.30, 0.35) both include zero, showing that the indirect effects of ties at both levels on relationship quality through communication are not significantly different from zero. Therefore, communication is not a mediator between interpersonal ties and relationship quality. For knowledge sharing, the bias-corrected bootstrap estimates (95% CI) of a1 (0.10, 0.94) and a2 (0.36, 4.91) exclude zero, indicating that both higher-level ties and lower-level ties are positively associated with knowledge sharing. The estimates of b1 (0.04, 0.60) also exclude zero, which means that knowledge sharing is related to relationship quality. In addition, the estimates of a1*b1 (0.02, 0.38) and a2*b2 (0.04, 1.10) both exclude zero, showing that the indirect effects of ties at both levels on relationship quality through knowledge sharing are significantly different from zero. Therefore, knowledge sharing is a mediator between interpersonal ties and relationship quality. Thus, H1a is partially supported.

As to conflict resolution, the bias-corrected bootstrap estimates (95% CI) of a1 (0.09, 1.07) and a2 (0.14, 0.82) exclude zero, indicating that both higher-level ties and lower-level ties are positively associated with conflict resolution. The estimates of b1 (0.03, 0.28) also exclude zero, which means that conflict resolution is related to relationship quality. In addition, the estimates of a1*b1 (0.03, 0.18) and a2*b2 (0.01, 0.13) both exclude zero, showing that the indirect effects of ties at both levels on relationship quality through conflict resolution are significantly different from zero. Thus, conflict resolution is a mediator between interpersonal ties and relationship quality. Similar to conflict resolution, the bias-corrected bootstrap estimates (95% CI) of a1 (0.01, 0.88) and a2 (0.42, 1.02) for cooperation exclude zero, indicating that both higher-level ties and lower-level ties are positively associated with cooperation. The estimates of b1 (0.06, 0.24) also exclude zero, which means that cooperation is related to relationship quality. In addition, the estimates of a1*b1 (0.01, 0.18) and a2*b2 (0.05, 0.19) both exclude zero, showing that for ties at both levels, the indirect effects on relationship quality through cooperation are different from zero. Cooperation is a mediator between interpersonal ties and relationship quality. Therefore, H1b is supported.

Next, we compare the effects of interpersonal ties at the higher and the lower levels. In doing so, we use path coefficients and standard errors from bootstrap results (see Table 5), and conduct coefficient comparisons through parametric *t* tests.[4] As shown in Table 5, the path coefficients of lower-level ties on all boundary-spanning behaviors are greater than those of higher-level ties. However, parametric *t* tests show that the path coefficients of lower-level ties on communication ($p < .05$), knowledge sharing ($p < .05$), and cooperation ($p < .01$) are significantly greater than those of higher-level ties, but the path coefficient of lower-level ties on conflict resolution is not significantly different from that of higher-level ties. Thus, H2a is fully supported and H2b is partially supported.

Finally, we test the interaction of higher-level ties and lower-level ties as stated in H3. As is shown in Table 4, the bias-corrected bootstrap estimates (95% CI) of a3 for communication (–0.17, 0.15) and knowledge sharing (–0.03, 0.69) both include zero, which means the interaction effects of interpersonal ties at both levels on communication and knowledge sharing are not significantly different from zero. Thus, H3a is not supported. Furthermore, the bias-corrected bootstrap estimates (95% CI) of a3 for conflict resolution (0.01, 0.26) and cooperation (0.04, 0.34) both exclude zero, which means the interaction effects of interpersonal ties at both levels on conflict resolution and cooperation are significantly different from zero, lending full support to H3b.

Table 5
Path Results With Bootstrap Methods (N = 1,000)

Path	Bootstrap *M*	Bootstrap *SE*	Parametric *t* Test
Interpersonal ties at the higher levels—Communication	0.36	0.51	56.53*
Interpersonal ties at the lower levels—Communication	10.21	0.53	
Interpersonal ties at the higher levels—Knowledge sharing	0.24	0.27	4.28*
Interpersonal ties at the lower levels—Knowledge sharing	10.21	0.50	
Interpersonal ties at the higher levels—Conflict resolution	0.55	0.22	1.72
Interpersonal ties at the lower levels—Conflict resolution	0.98	0.47	
Interpersonal ties at the higher levels—Cooperation	0.58	0.21	2.56*
Interpersonal ties at the lower levels—Cooperation	10.70	0.65	

*$p < .05$.

Comparing the results from the bootstrap method with the results from SEM and the traditional mediation analysis, we find that these results are largely consistent—the only exception is H1a. The discrepancies between the bootstrap results and the SEM results on knowledge sharing lead to partial support for H1a from the bootstrap results, and a rejection of H1a from the SEM results. A scrutiny of results, however, shows that while the links between interpersonal ties and knowledge sharing are signification from both methods, the link between knowledge sharing and relationship quality is significant (0.04, 0.60) in the bootstrap results, but is only marginally significant ($\beta = .12$, $p < .1$) in the SEM results. Taken together, it is reasonable to conclude that knowledge sharing is a mediator between interpersonal ties and relationship quality. Therefore, H1a is partially supported.

Discussion and Conclusion

Research on ties has attracted considerable interest from scholars. Previous studies have shown the importance of personal ties in driving superior performance, particularly for organizations in emerging markets. Our study advances extant literature by examining interpersonal ties in interorganizational exchanges, and by analyzing their link to the superior performance (i.e., relationship quality) of interorganizational relationships. As well, our study reveals the mediating forces through which strong interpersonal ties are related to favorable interorganizational relationships. After selecting China for the research context, we collected the matched dyadic data from manufacturer–distributor dyads in the Chinese household appliances industry in order to investigate interpersonal ties between boundary spanners in interorganizational exchanges. Overall, our study confirms the positive links between interpersonal ties and superior performance of interorganizational relationships, endorsing social embeddedness logic. By adding a boundary-spanning lens to dyadic exchange relationships, and by disentangling interpersonal ties to two different levels, our study offers new insights on the value of interpersonal ties in interorganizational exchanges in emerging markets.

A Boundary-Spanning View of Interpersonal Ties

Previous conceptual and empirical studies have focused on the micro–macro link between personal ties and performance. Drawing on social embeddedness theory, we extend our research on personal ties into interorganizational exchanges and examine the link between interpersonal ties and relationship quality of exchange relationships. Our results show that strong interpersonal ties are related to favorable exchange relationships that are characterized by a high degree of mutual trust, satisfaction, and commitment, and that indicate the significant value of interpersonal ties in achieving superior performance of exchange relationships in emerging markets. Thus, providing evidence from manufacturer–distributor relationships in China, this study concurs with prior studies that view ties as an indispensable strategic tool (Boisot & Child, 1996; Peng & Luo, 2000; Sheng et al., 2011), and confirms the significance of strong interpersonal ties in achieving favorable exchange relationships in emerging markets.

More important, however, are our findings on the mediating forces in the micro–macro link. The mediating forces are imperative to the understanding of interpersonal and interorganizational dynamics in exchanges, because the mediating forces help uncover the underlying mechanism through which interpersonal ties can be used to manage relationships at the interorganizational level. Adding a boundary-spanning lens to social embeddedness theory, this study contributes to ties research by providing a boundary-spanning mechanism that explains the value of interpersonal ties in the interorganizational dynamics of emerging markets. Boundary spanners in interorganizational exchanges are personnel who engage in exchange activities that represent exchange parties (Aldrich & Herker, 1977). A study of boundary spanners and their interpersonal ties would also highlight the important role of boundary spanners in interorganizational exchanges and in relationships management.

In general, the results demonstrate the mediating role of boundary-spanning behaviors in emerging markets, interceding between interpersonal ties and the relationship quality of exchange parties. Specifically, interpersonal ties between boundary spanners promote boundary-spanning behaviors—information processing (i.e., communication and knowledge sharing) and external representation (i.e., conflict resolution and cooperation). However, only knowledge sharing, conflict resolution, and cooperation help drive exchange parties' mutual trust, satisfaction, and commitment. In other words, communication is not a mediator between interpersonal ties and relationship quality. This finding implies that, although personal ties between boundary spanners improve communication, exchange organizations may take this behavior as a given in any exchange relationship, and therefore may not attach special meaning to the behaviors. In contrast, the boundary-spanning behaviors of knowledge sharing, conflict resolution, and cooperation are important to exchange organizations in emerging markets because those markets are typically characterized by less transparent information channels, as well as deficits in business knowledge and limitations in regulatory legal systems (Peng & Heath, 1996). The primary flow between organizations is information, and knowledge sharing guarantees the flow of information between exchange parties. Thus, knowledge sharing between exchange parties enhances the relationship quality of exchange relationships (Cheung et al., 2011). As well, for situations in which conflicts occur and cooperation is required, strong ties between boundary spanners can serve as a channel for boundary spanners to work together, resolving issues and getting things done (Wright et al., 2005).

These boundary-spanning behaviors in the form of external representation from both sides can effectively improve relationship quality for exchange parties in the focal exchange relationship.

A Two-Level Perspective of Personal Ties

Another interesting finding from this study is derived as a result of disentangling personal ties of two different but parallel levels in interorganizational exchanges. Extant studies on ties in emerging markets have largely focused on personal ties between top executives (e.g., Gu et al., 2008; Li et al., 2008; Peng & Luo, 2000), neglecting the fact that multiple levels of ties may exist in organizations. Our study, however, brings the importance of interpersonal ties at other levels to the forefront. We present an integrated model that investigates ties between boundary spanners at two levels—the higher level between top executives and the lower level between salespersons and buyers—in the interorganizational exchange. We also explore how interpersonal ties at the differentiated individual level may benefit relationship quality at the organizational level, solely and jointly. Specifically, our two-level approach to interpersonal ties offers three major findings that help to advance understanding of interpersonal ties in interorganizational exchanges in emerging markets.

First, our findings confirm that interpersonal ties do exist at two levels—the higher levels between executives and the lower levels between salespersons and buyers—across two organizations in an interorganizational exchange relationship. Both levels of interpersonal ties are positively associated with relationship quality that is mediated through boundary-spanning behaviors that include knowledge sharing, conflict resolution, and cooperation. In other words, knowledge sharing, conflict resolution, and cooperation are direct antecedents of relationship quality in interorganizational exchanges, while interpersonal ties are not. This finding is largely consistent with results from the relationship marketing literature (Palmatier et al., 2006).

Second, although both levels of interpersonal ties drive relationship quality through boundary-spanning behaviors and, thereby, facilitate dyadic perceptions of relationship quality, their effects are distinct. Our results suggest that for communication, knowledge sharing, and cooperation, the impetus of ties between salespersons and buyers is greater than that of ties between top executives. This finding suggests that, in interorganizational exchanges, lower-level ties are more effective than higher-level ties in enhancing mutual communication, knowledge sharing, and cooperation between exchange parties, whereas in emerging markets both levels of ties are equally effective in promoting conflict resolution. This finding not only extends our understanding of ties to encompass the distinctive effectiveness of different levels of personal ties but also implies the central role of lower-level boundary spanners in the interorganizational exchange relationships of emerging markets. Determining whether the same finding holds true in developed countries is an important empirical question that should be addressed.

Third, through examining the potential interaction between higher-level ties and lower-level ties, this study has uncovered a synergetic effect between two levels of interpersonal ties on driving boundary-spanning behaviors. Particularly, the positive effects of interpersonal ties on the facilitation of external presentation behaviors (i.e., conflict resolution and cooperation) are stronger when ties between top executives and ties between salespersons and buyers coexist. In contrast, the synergetic effects of interpersonal ties on promoting information processing behaviors (i.e., communication and knowledge sharing) are less salient. The fact that ties between top executives help organizations gain needed resources from partners and prevent partner opportunism (Westphal et al., 2006) in emerging markets may explain the amplifying effects of interpersonal ties on facilitating conflict resolution (conflicts due to partner opportunistic behaviors) and cooperation (when pooling resources is often required).

In summary, this study introduces a boundary-spanning mechanism and presents an integrative analysis of interpersonal ties in the dyadic interorganizational exchange of emerging markets. This approach to analysis of the issue is missing from the literature on ties. By focusing on two different levels of interpersonal ties, in view of their joint effects and relative power, we obtain answers to three aforementioned questions: (a) interpersonal ties are positively associated with relationship quality through boundary-spanning behaviors that include knowledge sharing, conflict resolution, and cooperation; (b) compared to interpersonal ties at the higher levels, ties at the lower levels have a larger association with relationship quality through communication, knowledge sharing, and cooperation; (c) interpersonal ties at the higher and the lower levels are complementary, and higher-level ties strengthen the associations between lower-level ties and external representation behaviors. As an investigation of the working mechanisms of two levels of interpersonal ties—between boundary spanners in interorganizational exchanges—this study enriches the literature on ties and interorganizational exchanges.

Managerial Implications

The results of this study provide several fruitful managerial implications for organizations in the interorganizational exchange of emerging markets. First, using dyadic data collected from manufacturer/supplier–distributor/buyer dyads in the Chinese household appliances industry, this study validates the importance of personal ties in China in general, and further suggests that interpersonal ties between boundary spanners is an effective relational governance tool for managing interorganizational relationships in emerging markets. Despite the potential side effects of personal ties in emerging markets (e.g., collective blindness, corruption) implied in several studies (Anderson & Jap, 2005; Gu et al., 2008), our research maintains that strong interpersonal ties cultivated between boundary spanners are associated with better relationship quality of exchange relationships. Thus, in opposition to Walmart’s prohibition against personal relationships between their buyers and suppliers’ salespersons in China, organizations are encouraged to avoid imposing such bans.

Second, this study further confirms that the positive influence of interpersonal ties on relationship quality works through boundary-spanning behaviors in exchange activities. This indirect link suggests that, in order to develop mutually perceived favorable relationships, exchange parties may choose either to encourage formation of ties between boundary-spanning personnel or to promote boundary-spanning behaviors—or both. With the latter option, for instance, organizations may encourage or even reward boundary spanners who are proactively involved in resolving conflicts and those who prevent conflicts from happening. If organizations choose to invest in interpersonal ties in order to ameliorate favorable relationships with exchange partners, organizations may encourage their boundary spanners to cultivate personal ties with their counterparts in partner organizations. This tactic is particularly useful in emerging markets (Luo, 2007; Peng & Heath, 1996) where many administrative loopholes and system imperfections exist because the legal systems in these markets have not kept up with the needs of economic development (Peng & Heath, 1996). Consequently, organizations must rely on interpersonal ties as an effective tool for developing and maintaining valuable relationships with exchange partners. For organizations, practical ways to create opportunities for nurturing interpersonal ties include establishing appropriate policies to encourage boundary spanners to spend time working together (Fock & Woo, 1998), providing social gatherings (e.g., holiday parties) and leisure activities (e.g., sports games) at which boundary spanners from exchange parties can participate (Li et al., 2008), or simply hiring as boundary spanners those persons who have already established ties and connections (Yang, 1994).

Third, although strong ties between boundary spanners are associated with high levels of boundary-spanning behaviors, only knowledge sharing, conflict resolution, and cooperation are positively associated with relationship quality. This finding suggests that communication between boundary spanners may be regarded as a norm by exchange parties in the interorganizational exchange of emerging markets; thus, communication may be treated by exchange parties as a necessary but insufficient condition for achieving favorable relationships with exchange partners. In order to develop mutually perceived favorable relationships, exchange parties need to steer boundary spanners toward greater focus on sharing explicit and tacit knowledge, resolving and preventing conflicts, and providing and inducing cooperative behaviors during interorganizational interactions and boundary-spanning activities. This is particularly true in emerging markets where market intelligence and knowledge are lacking, and where a large number of conflicts exist as a result of a weak legal system and incomplete contractual enforcement.

Fourth, this study examines interpersonal ties at two levels, and the results suggest that interpersonal ties between top executives, and between salespersons and buyers, generate a synergetic effect on dyadic relationship quality through conflict resolution and cooperation. As well, for the promotion of communication, knowledge sharing, cooperation, and relationship quality between exchange parties, the impact of ties between salespersons and buyers is greater than that of ties between top executives. In managerial applications, these findings imply that when sufficient resources are available, organizations should place emphasis on fostering ties between boundary spanners at both the higher and the lower levels, particularly when severe conflicts are expected, or when there is a need to cooperate in order to utilize resources in the focal exchange relationship. Such emphasis on fostering ties between boundary spanners (at both levels) will promote positive boundary-spanning interactions, and will improve, to the maximum degree, the relationship quality of the focal relationship. When under the constraints of limited resources, however, organizations should place the central focus on cultivating ties between boundary spanners at the lower levels, because this is the more effective method for achieving the desired favorable relationships with exchange partners.

Limitations and Future Research

Our results must be interpreted in light of the limitations of this study. First, despite the causal relationships implied by our proposed model (Figure 1) the cross-sectional data impose limits on our ability to draw causal inferences. Although research on ties suggests that personal socialization is an antecedent of subsequent business exchanges in emerging markets,[5] and though, on the basis of embeddedness logic, social embeddedness theory and boundary-spanning theory collectively theorize the causal links between interpersonal ties, boundary-spanning behaviors, and relationship quality, our cross-sectional data can prove only the associations among these constructs. To disclose the causal link, we followed up with recent interviews with one dozen senior managers and asked them for their view of the order of interpersonal ties and interorganizational relationships. The consensus that emerged showed that both interpersonal ties and interorganizational relationships are important to business, but that it is critical to establish interpersonal ties *ex ante*, which will subsequently nurture interorganizational relationships. In an effort to confirm the causal links and to reveal the causal effects of interpersonal ties in interorganizational exchanges, future studies may employ a longitudinal design and separate the data collection times in the order of interpersonal ties, boundary-spanning behaviors, and relationship quality, respectively.

Second, we use only data from China for testing the conceptual model and hypotheses, which may limit generalization of the findings to other countries. Although our conclusions may not be readily generalizable to developed nations, our key assumptions, theoretical model, and empirical findings are likely applicable to other emerging markets such as Russia, India, and Brazil, where interpersonal ties currently play a vital role in economic activities and business transactions (Peng & Heath, 1996). Future studies may collect data from other countries to allow for either validation of the model (if the data are from other emerging markets) or cross-country comparisons (if the data are from developed countries). Collecting data from other countries can also help refine and enrich the conceptual model. Interpersonal ties are complicated because, by definition, such personal relationships are based primarily on private personal feelings, and are generally subject to constraints from the external environment—culture, economic systems, or legal frameworks, for instance. Using data collected from multiple countries, future studies may add macro-level environmental factors and, in that context, may examine whether affiliations of interpersonal ties, boundary-spanning behaviors, and relationship quality are maintained under different environmental conditions.

Third, our data were collected from a single industry (i.e., the home appliance industry in China). This setting helps to eliminate industry-level variations that present potential noise in testing the model. Similar to the result of using only one country, however, using only one industry introduces industry-specific characteristics that may also limit the generalizability of the findings to other industries. As is seen in our sample setting, the long history and mature development of the home appliance industry in China has the result that many of the relationships between manufacturers and distributors in the home appliance industry are relatively longer than those in other industries. Undeniably, the average relationship duration in our sample is 5.44 years ($SD = 2.25$), which is longer than relationships in other industries (e.g., 3 years in Ambler et al., 1999). This might imply that interorganizational relationships in the home appliance industry are more harmonious than relationships in other industries in China. Indeed, despite the variance existing in each variable, the variables in our study show relatively high means (see descriptive statistics in Table 2). Future studies should collect data from other industries where the relationships are less stable in order to validate our proposed model.

Fourth, the control variables included in this study reflect characteristics at the organization and the relationship levels, overlooking the characteristics at the dyadic level that may affect relationship quality. For example, individuals from organizations with different ownership types may hold different perceptions toward conducting business transactions. Although ownership type was treated as a control variable in this study, a more meaningful approach might be to use as a control variable ownership congruence[6]—a dummy variable at the dyadic level indicating whether the manufacturer and the distributer in an exchange relationship have the same or different ownership types—because exchange parties with the same ownership type may develop a higher-quality relationship when a perception of legitimacy results from organizational structure isomorphism (DiMaggio & Powell, 1983). Furthermore, ownership congruence may be used as a moderator so that the link between interpersonal ties and relationship quality may be compared between domestic ownership dyads and foreign ownership dyads, with the goal of determining whether the effects of their tie utilizations vary by ownership (see Li et al., 2008). Another example of a dyadic control variable is power asymmetry—whether the exchange parties possess equal or differentiated power in the relationship. Taking the issues a step further, relationship characteristics at the dyadic level may also be investigated as moderating variables. For example, because state-owned firms are more likely to rely on personal relationships, the positive links between personal ties and relationship quality may be stronger when both parties are state-owned. Similarly, future studies may also consider collecting dyadic data in both symmetric and asymmetric power structures, in order to examine whether the proposed model holds under different power structures (e.g., the manufacturer power > the distributor power; the manufacturer power = the distributor power; the manufacturer power < the distributor power).

Fifth, although we include two levels of interpersonal ties and examine their interaction term, the dynamics between two levels of interpersonal ties remain unknown. It would be useful to consider whether interpersonal ties at the higher levels affect interpersonal ties at the lower levels, or vice versa. A further foundational question concerns how the dynamics of two levels of personal ties vary over the relationship lifecycle of interorganizational exchanges. Future studies may further the query by exploring the dynamics of interpersonal ties, at different levels and in different stages of the vibrant buyer–supplier relationship development process.

Finally, this study focuses on the outcome of personal ties between boundary spanners at two levels in interorganizational exchanges but does not analyze the determinants of interpersonal ties. With the view that interpersonal ties are endogenous rather than exogenous, expanded future studies may investigate the antecedents and the formation mechanisms of personal ties between top executives, and between purchasing agents and salespersons. Understanding the formation conditions of interpersonal ties at both levels would provide managers with clear guidelines for allocating resources in order to establish interpersonal ties at different levels, and for properly managing interpersonal ties afterward.

Notes

1. We deliberately selected 11 organizations of different sizes (large, medium, and small), different ownerships (state-owned, collective, private, wholly foreign-owned), and different geographical locations (coastal developed regions and relatively underdeveloped inlands) in order to rule out any potential influence from these factors. The interviewees were sales or purchasing managers, and they had been with their organizations 2 to 12 years, with an average of 6 years.

2. We also used the product of mean and interrater reliability as a proxy to test the model, and between the two approaches there are no differences in the overall model fit and hypothesis results.

3. The bootstrap method can calculate the indirect effect between each antecedent and the dependent variable in the mediation tests, while the traditional mediation analysis by Baron and Kenny (1986) can test only whether the proposed mediator is a mediator between all the antecedents and the dependent variables. Thus, the bootstrap method is recommended by Bollen and Stine (1990) over the traditional mediation analysis in the mediation tests when there are two or more antecedents in the model.

4. The coefficient comparison can be conducted by a parametric *t* test using the following formula,

$$t = \frac{b_i - b_j}{SE\, b_i - b_j}$$

where b_i represents the path coefficients of lower-level ties on one role behavior and b_j represents the path coefficients of higher-level ties on the same role behavior. The path coefficients and their standard errors from the bootstrap method are reported in Table 5. At the threshold value of $t\,(225\text{-}1)_{0.05} = 1.96$, we can conclude that lower-level ties have greater impact on relationship quality than do higher-level ties, through communication ($p < .05$), knowledge sharing ($p < .05$), and cooperation ($p < .05$), but not through conflict resolution ($p > .05$).

5. For example, Barden and Mitchell (2007) conclude that prior ties between two leaders (such as board interlock) increase the likelihood of subsequent exchanges between the two organizations.

6. We thank one anonymous reviewer for the suggestion of ownership congruence. Unfortunately, our sample did not allow us to include ownership congruence as a control variable because in our sample there were only 16 dyads of the same ownership type.

References

Adobor, H. 2006. The role of personal relationships in inter-firm alliances: Benefits, dysfunctions, and some suggestions. *Business Horizons*, 49: 473-486.

Aldrich, H., & Herker, D. 1977. Boundary spanning roles and organization structure. *Academy of Management Review*, 2: 217-230.

Ambler, T., Styles, C., & Wang, X. C. 1999. The effect of channel relationships and guanxi on the performance of inter-province export ventures in the People's Republic of China. *International Journal of Research in Marketing*, 16: 75-87.

Anderson, E., & Jap, S. D. 2005. The dark side of close relationships. *MIT Sloan Management Review*, 46: 75-82.

Anderson, E., & Weitz, B. 1992. The use of pledges to build and sustain commitment in distribution channels. *Journal of Marketing Research*, 29: 18-34.

Anderson, J. C., & Gerbing, D. W. 1988. Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, 103: 411-423.

Anderson, J. C., & Narus, J. A. 1990. A model of distributor firm and manufacture firm working partnerships. *Journal of Marketing*, 54: 42-58.

Bagozzi, R. P., & Yi, Y. 1988. On the evaluation of structural equation models. *Journal of the Academy of Marketing Science*, 16: 74-94.

Barden, J. Q., & Mitchell, W. 2007. Disentangling the influence of leaders' relational embeddedness on interorganizational exchange. *Academy of Management Journal*, 50: 1440-1461.

Baron, R. M., & Kenny, D. A. 1986. Moderator-mediator variables distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51: 1173-1182.

Boisot, M., & Child, J. 1996. From fiefs to clans and network capitalism: Explaining China's emerging economic order. *Administrative Science Quarterly*, 41: 600-628.

Bollen, K. A., & Stine, R. 1990. Direct and indirect effects: Classical and bootstrap estimates of variability. *Sociological Methodology*, 20: 115-140.

Cheung, M., Myers, M. B., & Mentzer, J. T. 2011. The value of relational learning in global buyer–supplier exchanges: A dyadic perspective and test of the pie-sharing premise. *Strategic Management Journal*, 32: 1061-1082.

DiMaggio, P., & Powell, W. W. 1983. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48: 147-160.

Droge, C., Claycomb, C., & Germain, R. 2003. Does knowledge mediate the effect of context on performance? Some initial evidence. *Decision Sciences*, 34: 541-568.

Efron, B., & Tibshirani, R. J. 1993. *An introduction to the bootstrap*. New York: Chapman and Hall.

Fock, H., & Woo, K. 1998. The China market: Strategic implications of guanxi. *Business Strategy Review*, 7: 33-44.

Fornell, C., & Larcker, D. F. 1981. Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 28: 39-50.

Foy, P. S. 1999. *Knowledge management in industry. Knowledge management handbook*. New York: CRC Press.

Friedman, R. A., & Podolny, J. 1992. Differentiation of boundary spanning roles: Labor negotiations and implications for role conflict. *Administrative Science Quarterly*, 37: 28-47.

Geyskens, I., & Steenkamp, J. E. M. 2000. Economic and social satisfaction: Measurement and relevance to marketing channel relationships. *Journal of Retailing*, 76: 11-32.

Granovetter, M. S. 1973. The strength of weak ties. *American Journal of Sociology*, 78: 1360-1380.

Granovetter, M. S. 1985. Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91: 481-510.

Griffith, D. A., Zeybek, A. Y., & O'Brien, M. 2001. Knowledge transfer as a means for relationship development: A Kazakhstan-foreign international joint venture illustration. *Journal of International Marketing*, 9: 1-18.

Gu, F. F., Hung, K., & Tse, D. K. 2008. When dose guanxi matter? Issues of capitalization and its dark sides. *Journal of Marketing*, 72: 12-28.

James, L. R., Demaree, R. G., & Wolf, G. 1984. Estimating within-group interrater reliability with and without response bias. *Journal of Applied Psychology*, 69: 85-98.

Kotabe, M., Martin, X., & Domoto, H. 2003. Gaining from vertical partnerships: Knowledge transfer, relationship duration, and supplier performance improvement in the U.S. and Japanese automotive industries. *Strategic Management Journal*, 24: 293-316.

Kumar, N., Scheer, L. K., & Steenkamp, J. E. M. 1995. The effects of perceived interdependence on dealer attitudes. *Journal of Marketing Research*, 32: 348-356.

Li, J. J., Poppo, L., & Zhou, K. Z. 2008. Do managerial ties in China always produce value? Competition, uncertainty, and domestic vs. foreign firms. *Strategic Management Journal*, 29: 383-400.

Luo, Y. 2007. *Guanxi and business* (2nd ed.). Singapore: World Scientific.

Marsden, P. V., & Campbell, K. E. 1984. Measuring tie strength. *Social Forces*, 63: 483-501.

Mohr, J. J., & Spekman, R. 1994. Characteristics of partnership success: Partnership attributes, communication behavior, and conflict resolution techniques. *Strategic Management Journal*, 15: 135-152.

Palmatier, R. W., Dant, R. P., & Grewal, D. 2007. A comparative longitudinal analysis of theoretical perspectives of interorganizational relationship performance. *Journal of Marketing*, 71: 172-194.

Palmatier, R. W., Dant, R. P., Grewal, D., & Evans, K. R. 2006. Factors influencing the effectiveness of relationship marketing: A meta-analysis. *Journal of Marketing*, 70: 136-153.

Park, S. H., & Luo, Y. 2001. Guanxi and organizational dynamics: Organizational networking in Chinese firms. *Strategic Management Journal*, 22: 455-477.

Peng, M. W., & Heath, P. 1996. The growth of the firm in planned economies in transition: Institutions, organizations, and strategic choice. *Academy of Management Review*, 21: 492-528.

Peng, M. W., & Luo, Y. D. 2000. Managerial ties and firm performance in a transition economy: The nature of a micro–macro link. *Academy of Management Journal*, 43: 486-501.

Podsakoff, P. M., & Organ, D. W. 1986. Self-reports in organizational research: Problems and prospects. *Journal of Management*, 12: 531-543.

Podsakoff, P. M., Mackenzie, S. B., Podsakoff, N. P., & Lee, J. Y. 2003. Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88: 879-903.

Sheng, S., Zhou, K. Z., & Li, J. J. 2011. The Effects of business and political ties on firm performance: Evidence from China. *Journal of Marketing*, 75: 1-15.

Stock, R. M. 2006. Interorganizational teams as boundary spanners between supplier and customer companies. *Journal of the Academy of Marketing Science*, 34: 588-599.

Su, C., Yang, Z., Zhuang, G., Zhou, N., & Dou, W. 2009. Interpersonal influence as an alternative channel communication behavior in emerging markets: The case of China. *Journal of International Business Studies*, 40: 668-689.

Tsang, E. W. K. 1998. Can guanxi be a source of sustained competitive advantage for doing business in China? *The Academy of Management Executive*, 12: 64-73.

Tushman, M. L., & Scanlan, T. J. 1981. Characteristics and external orientations of boundary spanning individuals. *Academy of Management Journal*, 24: 83-98.

Uzzi, B. 1997. Social structure and competition in interorganizational networks: The paradox of embeddedness. *Administrative Science Quarterly*, 42: 35-67.

Westphal, J. D., Boivie, S., & Chng, D. H. M. 2006. The strategic impetus for social network ties: Reconstituting broken CEO friendship ties. *Strategic Management Journal*, 27: 425-445.

Wright, M., Filatotchev, I., Hoskisson, E. E., & Peng, M. 2005. Strategy research in emerging economies: Challenging the conventional wisdom. *Journal of Management Studies*, 42: 1-33.

Xin, K., & Pearce, J. L. 1996. Guanxi: Connections as substitutes for formal institutional support, *Academy of Management Journal*, 39: 1641-1658.

Yan, L., Simsek, Z., & Lubatkin, M. H. 2008. Transformational leadership's role in promoting corporate entrepreneurship: Examining the CEO-TMT interface. *Academy of Management Journal*, 51: 557-576.

Yang, M. M. 1994. *Gifts, favors and banquets*. New York: Wilder House Series, Cornell.

Zhang, J., Soh, P., & Wong, P. 2010. Entrepreneurial resource acquisition through indirect ties: Compensatory effects of prior knowledge. *Journal of Management*, 36: 511-536.

Zhang, Y., & Zhang, Z. 2006. Guanxi and organizational dynamics in China: A link between individual and organizational levels. *Journal of Business Ethics*, 67: 375-392.

Zollo, M., Reuer, J. J., & Singh, H. 2002. Interorganizational routines and performance in strategic alliance. *Organization Science*, 13: 701-713.